Velocity Verlet Stability

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Velocity varlet algorithm for numerical integration of equation of motion (namely $F = ma$) is ubiquitous in molecular dynamics. In this note we look at the upper limit of the size of one time step, so we can choose a large step to speed up simulation, but it should not be so large that the integrator blows up.

First we define a few things. System at at real time t_n corresponds to numerical propagation after n steps, $t_n = n\Delta t$ where the step size of each step is Δt. Then similarly the position at time t_n is $q_n = q(t_n)$, momentum is $p_n = p(t_n)$ and force is $f_n = f(t_n)$ The velocity algorithm consists of 3 steps

$$p_{n+1/2} = p_n + f_n \frac{\Delta t}{2} \tag{1}$$

$$q_{n+1} = q_n + \frac{p_{n+1/2}}{m}\Delta t \tag{2}$$

$$p_{n+1} = p_{n+1/2} + f_{n+1}\frac{\Delta t}{2} \tag{3}$$

substitute Eq.(1) into Eq.(2) yields

$$q_{n+1} = q_n + \dot{q}_n \Delta t + \frac{1}{2}\ddot{q}_n \Delta t^2 \tag{4}$$

which is actually just one of the SUVAT equations for constant acceleration we learnt back in school, namely $s = ut + \frac{1}{2}at^2$ (s is displacement, u is initial velocity v is final velocity a is acceleration an t is time elapse). Substitute Eq.(1) into Eq.(3) yields

$$p_{n+1} = p_n + \frac{f_n + f_{n+1}}{2}\Delta t \tag{5}$$

so momentum is just propagated by the average force of the current step and the next step in velocity Verlet. We divide through by the mass m on both sides of Eq.(5) yields

$$\dot{q}_{n+1} = \dot{q}_n + \frac{\ddot{q}_n + \ddot{q}_{n+1}}{2}\Delta t \tag{6}$$

Another SUVAT equation we should know from school is $s = vt - \frac{1}{2}at^2$ which is also very easy to derive by considering $n = n' - 1$ in Eq.(4) and (5)

$$q_n = q_{n-1} + \dot{q}_n \Delta t - \frac{1}{2}\ddot{q}_n \Delta t^2 \tag{7}$$

rearrange for $\dot{q}_n \Delta t$ then substitute back into Eq.(4) gives

$$q_{n+1} - 2q_n + q_{n-1} = \ddot{q}_n \Delta t^2 \tag{8}$$

For a harmonic oscillator

$$\ddot{q}_n = -\omega^2 q_n \tag{9}$$

substitute this into Eq.(8) gives

$$q_{n+1} - 2q_n + q_{n-1} = -(\omega\Delta t)^2 q_n \tag{10}$$

this is a recurrence relation that requires 2 initial values (like a Fibonacci sequence). Furthemore, it is linear, i.e. any linear combination of the valid solutions is still a valid solution. Lets try to solve it by using the substitution $x_n = \lambda^n$, then we get a quadratic equation in λ

$$\lambda^2 - (2 - (\omega\Delta t)^2)\lambda + 1 = 0 \tag{11}$$

after solving for the two roots λ_+, λ_-, we have $q_n = c_+\lambda_+^n + c_-\lambda_-^n$, where c_+ and c_- are coefficients, which can be found by solving the simultaneous equations

$$q_0 = c_+ + c_-, \qquad q_1 = c_+\lambda_+ + c_-\lambda_- \tag{12}$$

from Eq.(11), we must have $\lambda_+\lambda_- = 1$, but if any $|\lambda_\pm| > 1$, one of the term in Eq.(13) will blow up, causing the position also to blow up, to avoid this, we set $\lambda_\pm = e^{\pm i\theta}$, then we have

$$2\cos\theta = 2 - (\omega\Delta t)^2 \tag{13}$$

the LHS is oscillating between -2 and $+2$, so the RHS must satisfy

$$0 \leq (\omega\Delta t)^2 \leq 4 \tag{14}$$

$$0 \leq \omega\Delta t \leq 2 \tag{15}$$

Now consider the edge case of $\omega\Delta t = 2$, substitute this back into Eq.(11) gives repeated roots $\lambda_\pm = -1$. Now let $q_n = y_n(-1)^n$, which is not restrictive as any array can be written in this way by adding a minus sign every other entry, substitute this back into Eq.(10) gives

$$y_n - y_{n-1} = y_{n+1} - y_n \tag{16}$$

so y_n increase linearly with n, and q_n oscillates between $+$ and $-$ whilst its magnitude increases linearly, so this is also not stable. Thus, we have a rather clean criterion on the stability of the integrator in terms of the size of integration time step, for a harmonic oscillator.

$$\boxed{\Delta t < \frac{2}{\omega}} \tag{17}$$